Mathematical Foundations and Practical Interpretation of Statistical Inference: From Hypothesis Testing on Means to Linear Predictive Models

NECKER

June 3, 2026

Contents

1 Hypothesis Testing: The Decision-Making Process

Hypothesis testing is a statistical procedure that allows making decisions about a population based on information provided by a sample. The core logic is assimilable to that of a court trial, where the principle of presumption of innocence applies.

1.1 The Hypotheses and the Significance Level (α)

The test is set up by counterposing two hypotheses:

- **Null Hypothesis (H_0):** Represents the status quo, the presumption of innocence. It usually indicates the absence of an effect, equality, or the maintenance of a declared standard.
- **Alternative Hypothesis (H_1):** Represents the claim one wishes to prove (the prosecution).

The significance level, denoted by α (Alpha), **is not a physical tolerance** or an industrial one (e.g., a margin of error in grams), but rather a tolerance on the *risk of making a wrong decision*. Specifically, α is the maximum tolerated probability of committing a **Type I Error** (false positive): rejecting H_0 when it is actually true. Setting $\alpha = 0.05$ (5%), for instance, means accepting a maximum 5% probability of wrongfully convicting an innocent null hypothesis due to an unfortunate sample.

1.2 Test Statistic and p-value

Once data is collected, a **Test Statistic** (e.g., Z or t) is calculated. This converts raw data into a standardized distance from the null hypothesis.

The **p-value** is the probability, computed assuming H_0 is true, of obtaining a result equal to or more extreme than the observed one. It is the actual risk of a false alarm inherent in the data.

- If **p-value** $< \alpha$: The event is too rare to be a result of pure chance. One rejects H_0.
- If **p-value** $\geq \alpha$: There is insufficient evidence (beyond reasonable doubt) to reject H_0.

Deep Dive: Finding the Critical Z Value

The Critical Value is the translation of α onto the abscissa axis of the standardized distribution. It defines the boundary of the rejection region. Without advanced calculators, for a one-tailed test with $\alpha = 0.05$, one searches within the normal distribution tables for the probability values closest to 0.0500. Since the value lies exactly halfway between 0.0505 ($Z = -1.64$) and 0.0495 ($Z = -1.65$), interpolation yields the critical value of $\mathbf{-1.645}$.

WARNING: the Gaussian curve represents the ideal situation, therefore if **p-value** $< \alpha$, it means we are in a highly improbable zone and that the hypothesis H_0 does not faithfully reflect the total population.

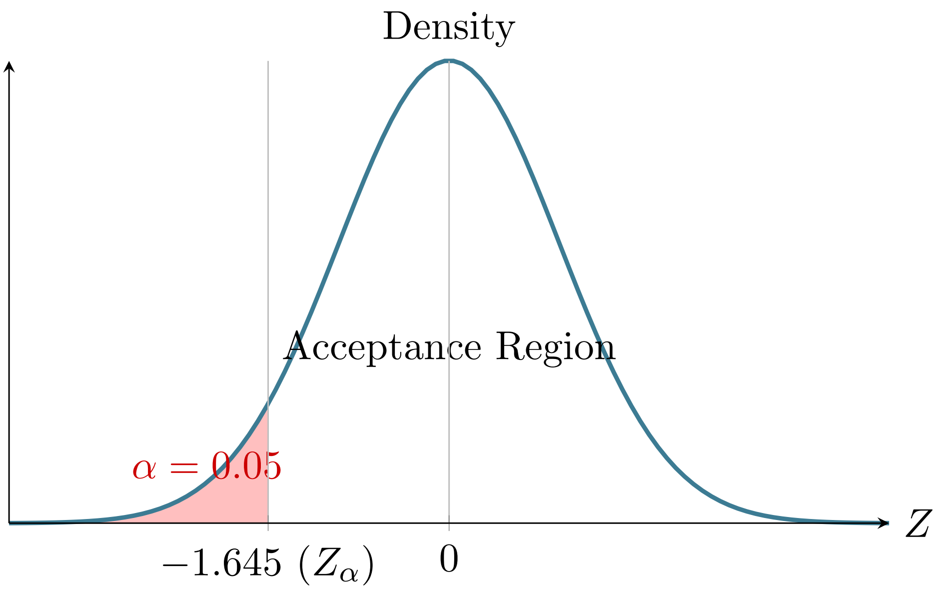


Figure 1: Representation of the rejection region (left tail) for $\alpha = 0.05$.

1.3 Application Example: Verifying the Mean Content

A manufacturer claims $\mu = 500$g. A sample of $n = 36$ is drawn, yielding a sample mean $\bar{x} = 495$g. The known standard deviation is $\sigma = 12$g (of the total population, found by estimation). **We want to check if the probability of extracting a box of 495g or lower, if the average is truly 500g per box, is less than the maximum accepted probability ($\alpha = 0.05$).**

- $H_0 : \mu = 500$ against $H_1 : \mu < 500$
- Test Statistic: $Z = \frac{495-500}{12/\sqrt{36}} = -2.5$
- Critical Value $Z_\alpha = -1.645$
- **Conclusion:** Since $Z = -2.5$ falls into the rejection region (beyond -1.645), H_0 is rejected. The associated p-value is 0.0062 (0.62%), which is strictly lower than 5%.

2 Choosing the Distribution: Standard Normal (Z) vs. Student's t

Should we use the **Standard Normal (Z)** distribution or the **Student's** t distribution? The choice is not arbitrary, but strictly depends on the information available to us regarding the population and the size of the sample.

2.1 The Problem of Additional Uncertainty

The Standard Normal distribution is the "perfect" model. It is used when we know exactly the variability of the entire population, meaning its standard deviation (σ).

However, in empirical reality, we almost never know the variance of the entire population. If we are measuring the weight of a sample of packages, we know the standard deviation *of our sample* (s), not that of all packages ever produced (σ). Substituting σ with s introduces a **new tier of uncertainty**: not only might our sample mean be imprecise, but now our error estimation might be as well.

This is where the **Student's** t distribution comes into play (pseudonym of the statistician William Sealy Gosset). The t distribution is mathematically designed to "absorb" this additional uncertainty.

2.2 Geometric and Theoretical Differences

Both the Normal (Gaussian) curve and the Student's t curve share some fundamental properties:

- They are both bell-shaped (campanulate).
- They are both symmetric with respect to zero (mean = mode = median = 0).

The crucial differences lie in the **shape of the tails**:

- **The tails of the Student's t are "heavier" (thicker):** Since there is more uncertainty (due to using s instead of σ), the probability of obtaining extreme values or values far from the mean is greater compared to a perfect Normal distribution. Consequently, the critical values of t are numerically larger than those of Z for the same α level, making the tests more "conservative" and the confidence intervals wider.
- **Dependence on Degrees of Freedom (df):** Unlike the Standard Normal, which is a single fixed curve, Student's t is actually a *family of curves*. The exact shape depends on the Degrees of Freedom ($df = n - 1$). The larger the sample (and therefore the more the sample standard deviation becomes a reliable estimate of σ), the more the degrees of freedom increase.
- **Convergence:** By the Law of Large Numbers, as $n \to \infty$ (conventionally for $n > 30$), the Student's t curve thins out and becomes practically indistinguishable from the Standard Normal distribution.

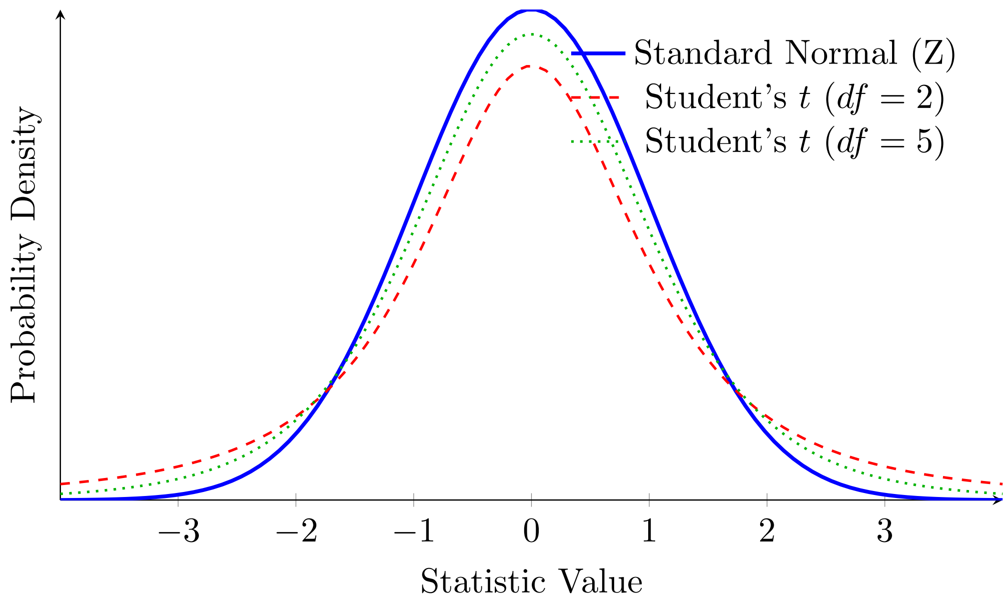


Figure 2: Comparison between the Normal and Student's t distributions. Notice how the t curves have a lower peak and "thicker" tails, implying a higher probability of extreme values due to sampling uncertainty. As the degrees of freedom increase, the green curve approaches the blue one.

2.3 How and When to Use Them

The computational procedure is identical for both: a test statistic is calculated and compared with a critical value (or the p-value is computed). Only the reference table and the letter used for the formula change.

Operational steps to use the Student's t:

1. **Verification of the requirement:** Ensure that σ is **unknown** and that you are using the sample standard deviation (s).

2. **Calculation of degrees of freedom:** For a test on the mean of a single sample, $df = n - 1$. WARNING: that one can become a higher number depending on the number of unknowns to estimate EG: parameters of a straight line y = mx + q (2)

3. **Test Statistic:** Substitute σ with s. The formula becomes:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

4. **Statistical Tables:** Enter the Student's t tables crossing the newly calculated degrees of freedom (df) with the desired significance level (e.g., $\alpha = 0.05$).

Decision Matrix

To avoid confusion on the exam, here is the simplified classic decision rule:

Conditions	**Small Sample** ($n \leq 30$)	**Large Sample** ($n > 30$)
Population σ KNOWN	Normal (Z)	Normal (Z)
σ UNKNOWN (using s)	Student's t	*Formally* t (Often approximated to Z)

3 When to Use Whole α vs. $\alpha/2$

3.1 Using Whole α: One-Tailed Tests (Unilateral)

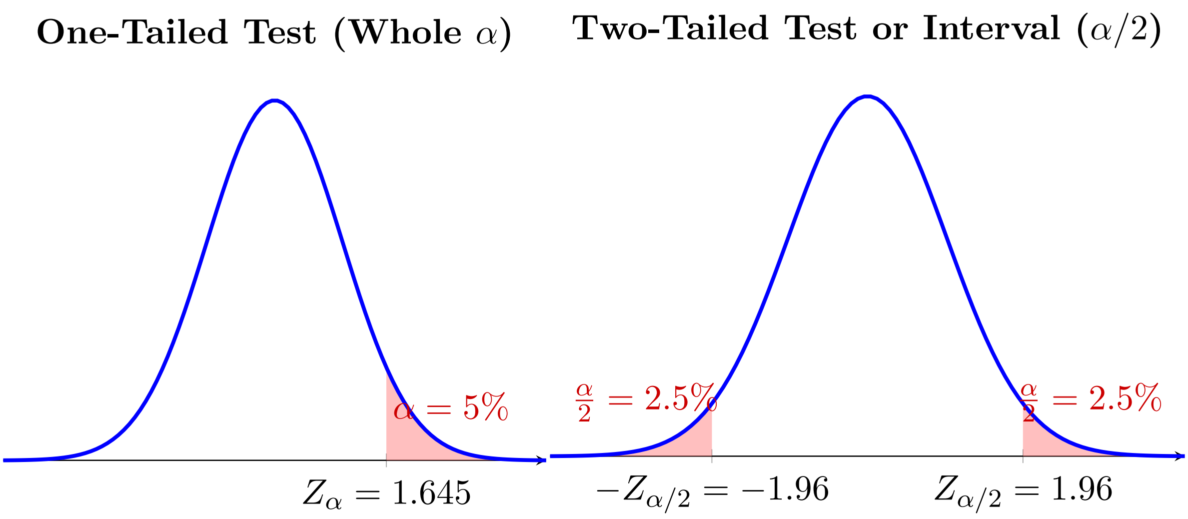
The entire error α (e.g., 5%) is concentrated on a single side of the distribution (either only the right tail or only the left tail) when the Alternative Hypothesis (H_1) has a very precise direction.

- **Keywords in text:** "Greater than", "Less than", "Improved", "Lower".
- **Logic and Application:** In our first exercise, the suspicion was that the weight was *less* than 500g. We did not care at all if the packages accidentally weighed more (that would be a loss for the company, not for us consumers!). Since we are only interested in one direction, we concentrate all our "investigative power" (the 5% accepted risk) on the left tail.

3.2 Using $\alpha/2$: Two-Tailed Tests and Intervals (Bilateral)

The error is split exactly in half ($\alpha/2$ on the right and $\alpha/2$ on the left) in two fundamental situations of inferential statistics:

- **Bilateral Hypothesis Tests:** When we are interested in discovering any difference, in any direction (Keywords: "Different from", "Changed", "Has an effect"). In this case, the alternative hypothesis H_1 uses the $\neq$ symbol. Since the anomaly could manifest either in excess or in defect, we must divide our flank equally (e.g., 2.5% risk on the right and 2.5% on the left).



4 Simple Linear Regression

While hypothesis tests evaluate assertions, simple linear regression aims to mathematically model the relationship between an independent variable (X, explanatory) and a dependent variable (Y, response), to understand its impact and make predictions. The theoretical equation is $\hat{Y} = \beta_0 + \beta_1 X$.

4.1 The Model Coefficients (β_0 and β_1)

The objective is to draw the line that minimizes the distance from the real data.

- **Slope (β_1):** It is calculated as the ratio between the covariance of the two variables and the variance of the independent variable (SS_{XY}/SS_{XX}). *Practical interpretation:* Indicates the expected mean change in Y for each unit increment in X.

- **Intercept (β_0):** It is calculated by forcing the line to pass through the centroid of the data ($\bar{X}, \bar{Y}$). The inverse formula is
$\beta_0 = \bar{Y} - \beta_1\bar{X}$. *Practical interpretation:* It is the theoretical expected value of Y when $X = 0$.

4.2 Goodness of Fit: The Coefficient of Determination (R^2)

The R^2 is the model's "report card", a value bounded between 0 and 1 (or 0% and 100%). It quantifies the proportion of variance in the dependent variable Y that is explained by the linear model through the variable X. A very high R^2 (e.g., > 0.90) indicates that the data aligns almost perfectly along the line.

The **adjusted** R^2 introduces a penalty depending on the number of independent variables included, preventing the statistical illusion that a model improves simply by adding useless variables at random.

Deep Dive: Degrees of Freedom ($n - k - 1$)

In statistics, degrees of freedom represent the "budget of independent pieces of information" available. For each parameter that must necessarily be estimated from the data before the model's error can be computed, a mathematical constraint is introduced and a degree of freedom is lost.

- In a test on a single mean, 1 parameter is estimated ($\bar{x}$), hence $df = n - 1$.

- In simple regression, to draw the line, 2 parameters must be estimated (β_0 and β_1), so the error on residuals is computed with $df = n - 2$.

4.3 Uncertainty: Confidence Intervals vs. Prediction Intervals

Once the coefficients are computed, it is necessary to evaluate how much the estimated model can be trusted compared to the true and unknown total population. To do so, one abandons the single point estimate (which is by nature almost certainly imprecise) and constructs a logical "safety enclosure". All intervals follow the same universal structure: **Point Estimate $\pm$ (Critical Value $t\times$ Standard Error)**.

There are two fundamental types of intervals applied to the regression model:

1. **Confidence Interval for the slope (β_1):** Evaluates the structural uncertainty of the model, i.e., the global effect of X on Y.

    - *The deep meaning:* We have calculated a slope based on a single and limited sample of n data points. If we could repeat the experiment infinite times, extracting infinite different samples, we would get lines that are always slightly different. This interval tells us that we are confident (e.g., at 95%) that the "true" and invisible universal slope of the population falls within these bounds. Above all, if the interval does *not* include zero, it means that the effect of the variable X is real, guaranteeing its statistical significance.

- *How it is calculated:* The standard error of the slope $SE(\beta_1)$ is used, which depends on how scattered the points are around the line (the residual error s_e) and how large the variance of X is (SS_{XX}). The formula is:

$$\beta_1 \pm \left(t_{\alpha/2} \times \frac{s_e}{\sqrt{SS_{XX}}} \right)$$

2. **Prediction Interval:** Used when one wants to estimate not the mean effect, but the exact value of Y for a *single and specific future event* given a certain value X_0.

   - *The deep meaning:* Predicting the outcome of a single event is inherently harder and more uncertain than predicting a global average (for example, a single agricultural week can suffer an unexpected hailstorm). For this reason, the prediction "enclosure" must be mathematically wider to absorb the high natural uncertainty of the single contingent case.
   - *How it is calculated:* Within the standard error of the prediction (SE_{prev}), an extra dispersion factor is added (the "+1" under the square root). The complete formula is:

$$\hat{Y}_0 \pm \left(t_{\alpha/2} \times s_e \sqrt{1 + \frac{1}{n} + \frac{(X_0 - \bar{X})^2}{SS_{XX}}} \right)$$

   Crucial note: The presence of the term $(X_0 - \bar{X})^2$ under the root indicates that the error grows as the chosen value X_0 deviates further from the mean $\bar{X}$. Mathematically, this means that the model is very secure when making predictions close to the center of mass of past data, but becomes increasingly uncertain (and the interval widens like a funnel) if we try to use it for extreme or never-before-observed values of X.

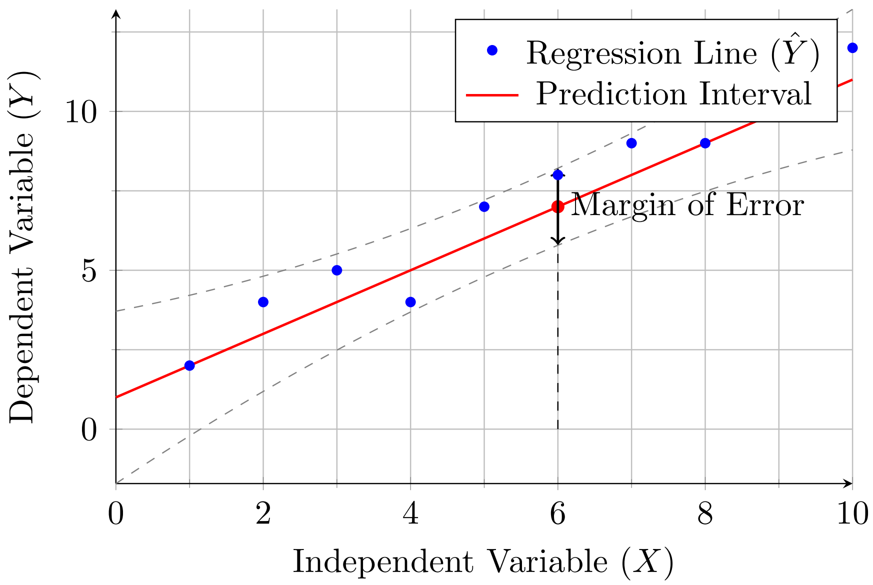


Figure 3: Linear model with dispersion of real data and prediction bands for a single event. Notice how the dashed bands widen slightly as they move away from the center of the data.

4.4 Application Example: Agricultural Yield (Fertilizer and Tomatoes)

Aggregated data for $n = 10$ weeks (**X** indicates the kg of fertilizer used, **Y** indicates the kg of tomatoes produced):
$SS_{XX} = 82.5$, $SS_{YY} = 88.4$, $SS_{XY} = 83$.
Means: $\bar{X} = 5.5$, $\bar{Y} = 6.6$.

- **Calculation of Coefficients:**
  Applying the theoretical formulas for the slope (β_1) and the intercept (β_0):

$$\beta_1 = \frac{SS_{XY}}{SS_{XX}} = \frac{83}{82.5} \approx 1.006$$

$$\beta_0 = \bar{Y} - \beta_1 \bar{X} = 6.6 - (1.006 \times 5.5) \approx 1.067$$

  The estimated model is therefore: $\hat{Y} = 1.067 + 1.006X$.

- **Single Prediction for a new value $X_0 = 6$ kg:**
  First, we calculate the **point estimate** by inserting the value into the model:

$$\hat{Y} = \beta_0 + \beta_1 X_0 = 1.067 + 1.006(6) \approx 7.103 \text{ kg}$$

  Due to the variability of a single event (e.g., weather conditions during that specific week), the point estimate is not enough and we must construct a **95% Prediction Interval**.

  First, we calculate the Sum of Squared Errors (SSE) and the standard error of the residuals (s_e):

$$SSE = SS_{YY} - \beta_1 SS_{XY} = 88.4 - (1.006 \times 83) \approx 4.897$$

$$s_e = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{4.897}{8}} \approx 0.782$$

  Now we compute the augmented standard error for predicting a single event (SE_{prev}):

$$SE_{prev} = s_e \sqrt{1 + \frac{1}{n} + \frac{(X_0 - \bar{X})^2}{SS_{XX}}}$$

$$SE_{prev} = 0.782 \sqrt{1 + \frac{1}{10} + \frac{(6 - 5.5)^2}{82.5}} \approx 0.822$$

  Finally, knowing that the Student's t critical value for a two-tailed test with $\alpha = 0.05$ and $df = 8$ is 2.306, we assemble the interval:

$$\hat{Y} \pm (t_{\alpha/2} \times SE_{prev})$$

$$7.103 \pm (2.306 \times 0.822) \implies 7.103 \pm 1.895$$

  We obtain the prediction interval $[5.208, 8.998]$. We are 95

  WARNING: 95% reliability means we must widen the boundaries of the slope enough to have a 95% probability of landing on a line that has a coefficient within the established boundaries.